



***Functional Outcome of Cervical Spondylotic Myelopathy (CSM) Patients at
Prof. Soeharso Orthopedic Hospital: A Retrospective Study***

Introduction



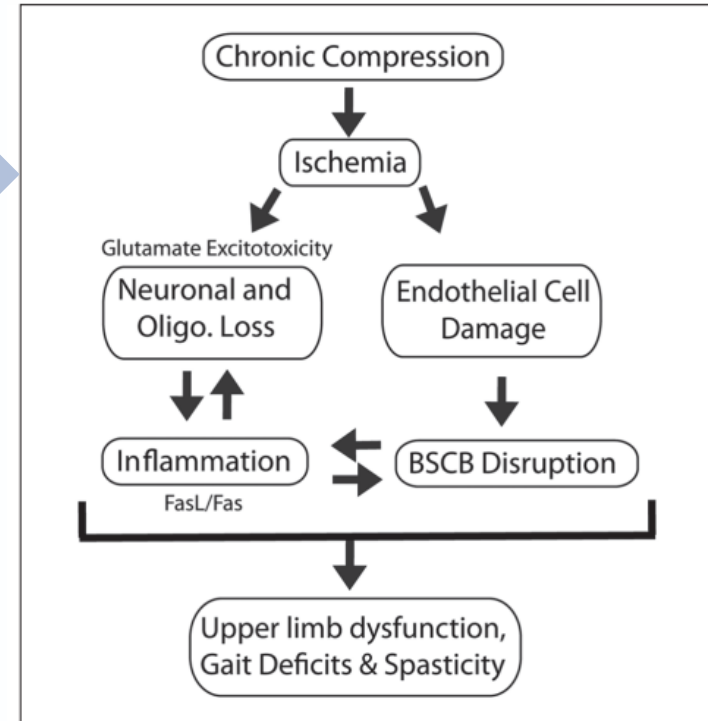
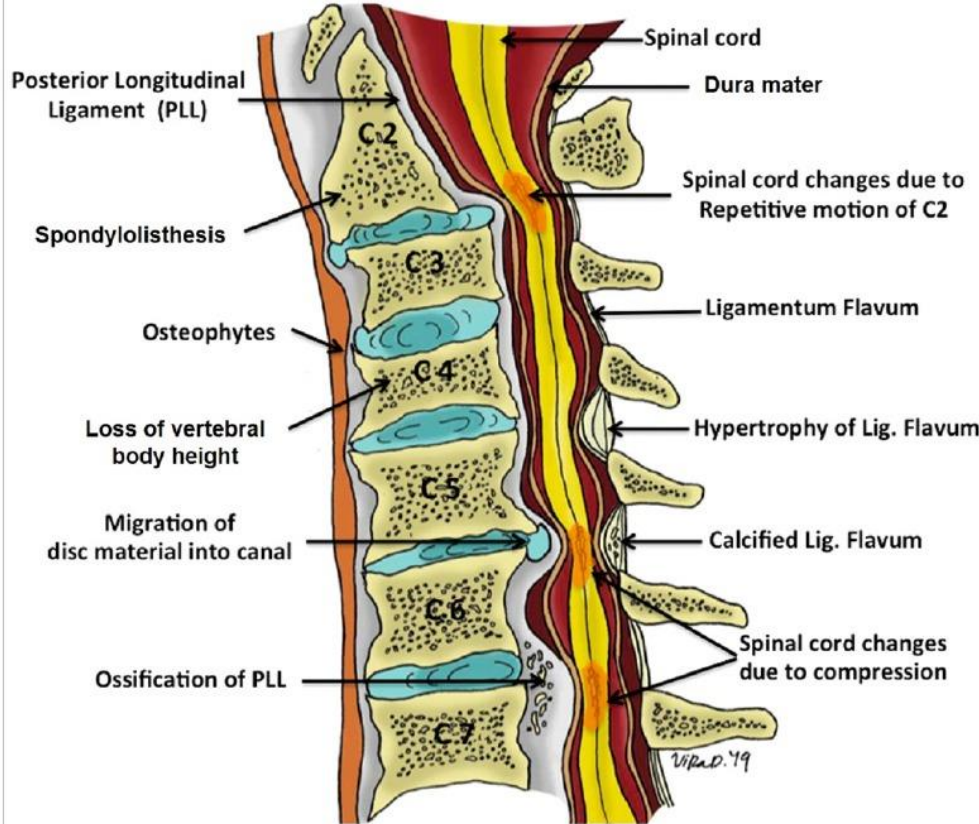
- Cervical Spondylotic Myelopathy (CSM) is an altered neurological condition due to a degenerative process that results in compression of the spinal cord and surrounding structures.
- Asia → CSM incidence 4.04 per 100,000 people/year

Introduction



- Functional scales used to assess the severity of CSM patients, which include the Nurick scale, and the modified Japanese Orthopaedic Association (JOA) scale.
- X Ray parameters on the Cervical one of which is used is cSVA (*Cervical Sagittal Vertical Axis*)
- MRI is used to determine the location of the lesion. MRI can depict pathology such as disc herniation and facet hypertrophy.
- Management of CSM can be conservative or surgical.
- Surgical options include anterior, posterior or combined approaches.

Pathogenesis of CSM



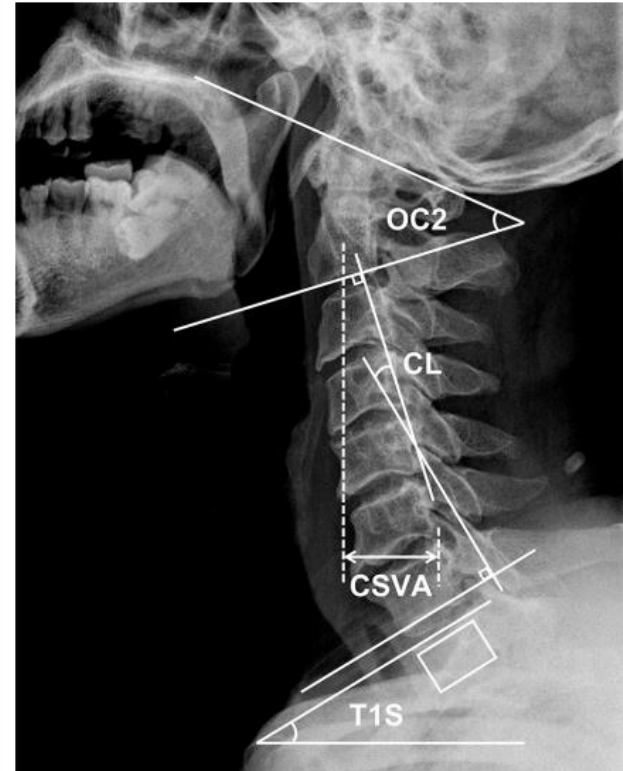
mJOA Scoring

- Mild myelopathy: 15-17
- Moderate myelopathy: 12-14
- Severe Myelopathy: 0-11

Category	Score	Description
Upper Extremity Motor Subscore (/5)	0	Unable to move hands
	1	Unable to eat with a spoon but able to move hands
	2	Unable to button a shirt but able to eat with a spoon
	3	Able to button a shirt with great difficulty
	4	Able to button a shirt with mild difficult OR other mild fine motor dysfunction (marked handwriting change, frequent dropping of objects, difficult clasping jewelry, etc.)
	5	Normal hand coordination
Lower Extremity Subscore (/7)	0	Complete loss of movement and sensation
	1	Complete loss of movement, some sensation present
	2	Inability to walk but some movement
	3	Able to walk on flat ground with walking aid
	4	Able to walk without walking aid, but must hold a handrail on stairs
	5	Moderate to severe walking imbalance but able to perform stairs without handrail
	6	Mild imbalance when standing OR walking
Upper Extremity Sensory Subscore (/3)	7	Normal walking
	0	Complete loss of hand sensation
	1	Severe loss of hand sensation OR pain
	2	Mild loss of hand sensation
Urinary Function Subscore (/3)	3	Normal hand sensation
	0	Inability to urinate voluntarily (requiring catheterization)
	1	Frequent urinary incontinence (more than once per month)
	2	Urinary urgency OR occasional stress incontinence (less than once per month)
	3	Normal urinary function

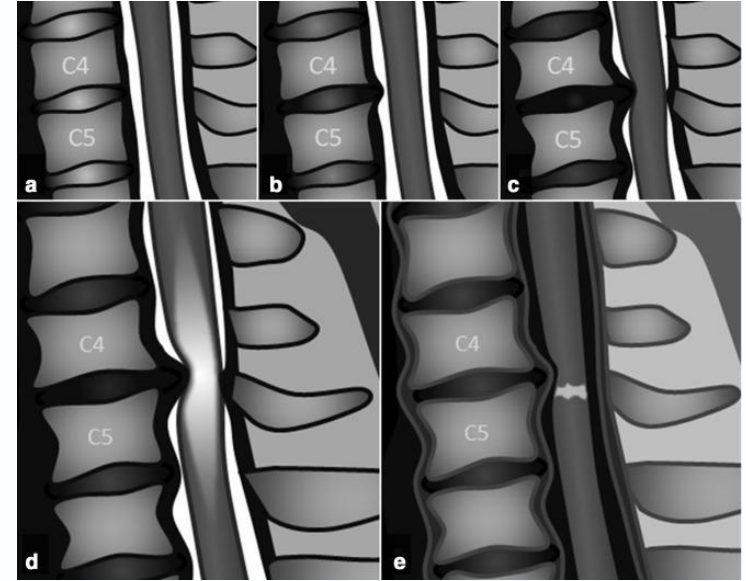
Diagnosis

- On X Ray, Cervical abnormality can be measured *by Cervical Sagittal Vertical Axis (cSVA)*.
- cSVA was measured as the distance between the perpendicular plumb line of C2 to the posterosuperior of the upper end plate of C7 in millimeters (mm).
- Interpretation is negative (-) if the line is found behind the superoposterior C7 and positive (+) if the plumb line falls behind the posterosuperior C7.
- Xu et al → Preoperative CSVA is significantly associated with neurologic outcomes after laminoplasty



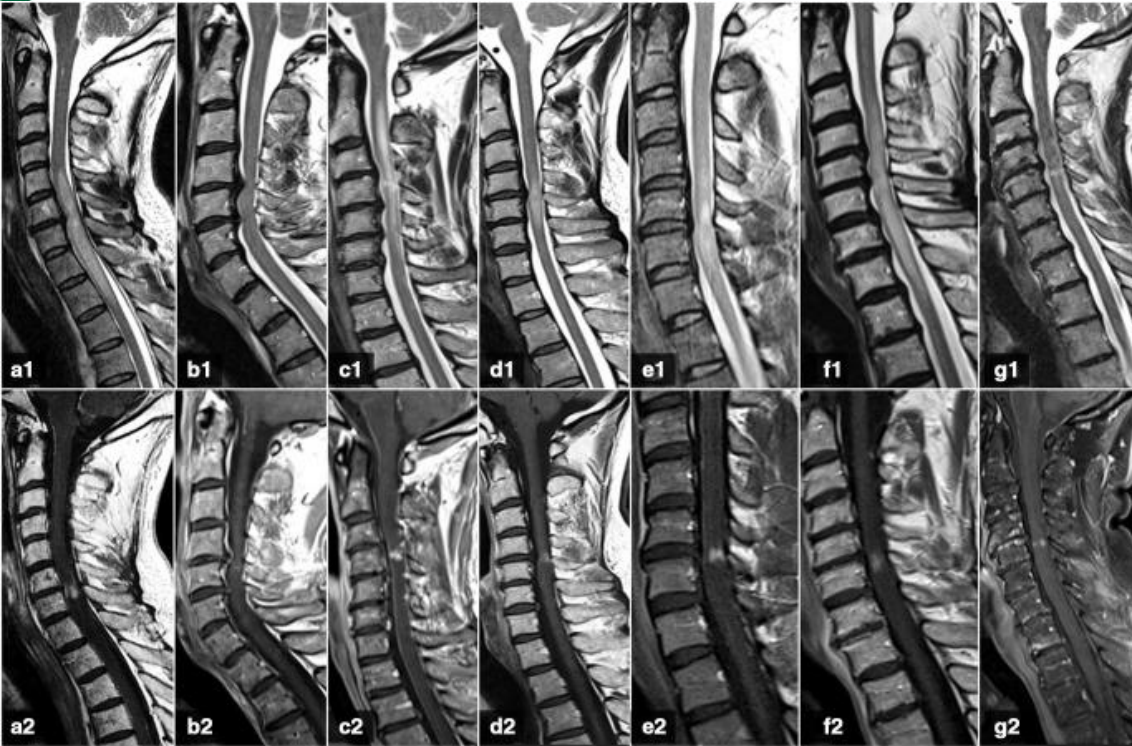
Diagnosis

- MRI provides excellent anatomical information on the macro-structure of the spinal cord
- Central stenosis is often characterized by:
 - Degenerative hypertrophy of the facet joints
 - Osteophytes
 - Thickening of the ligamentum flavum
 - Ossification of the posterior longitudinal ligament
 - Posterior intervertebral disc protrusion



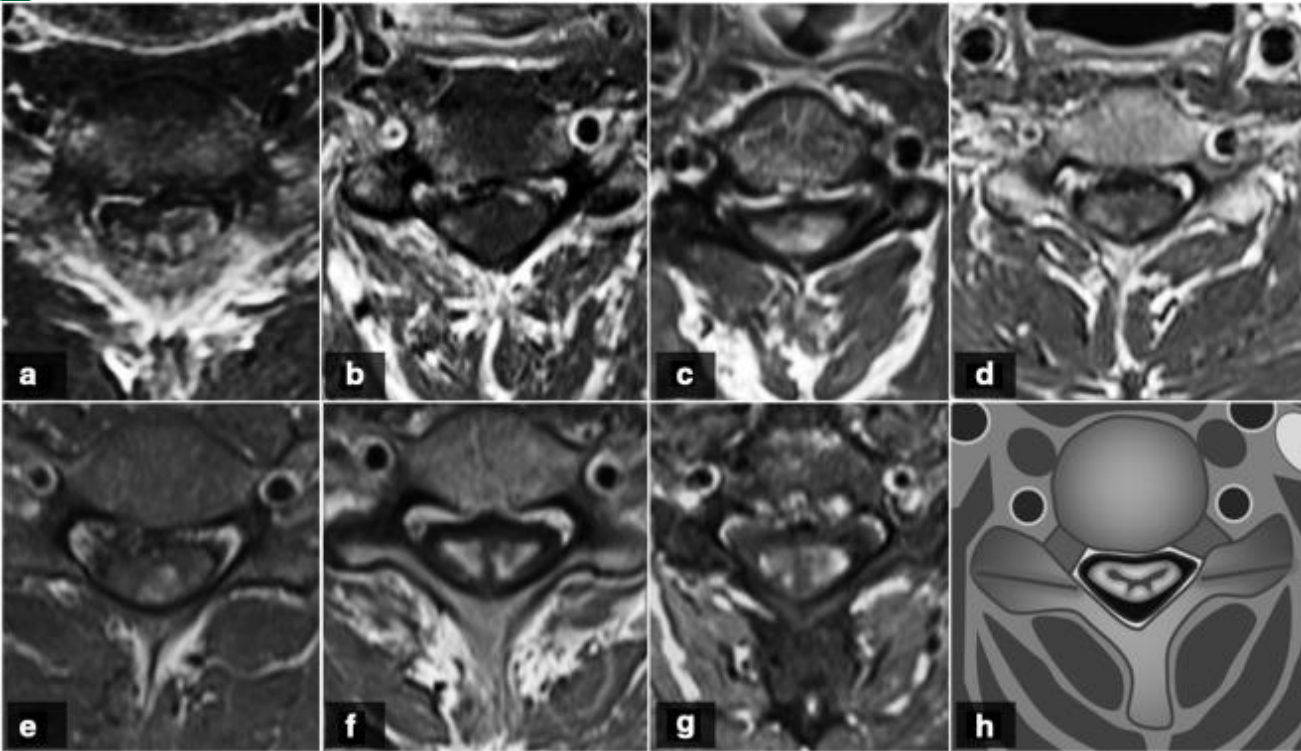
MRI grading by Kang et al on sagittal T2-WI:

(a) Grade 0, absence of canal stenosis; (b) Grade 1, obliteration of subarachnoid space exceeding 50%; (c) Grade 2, spinal deformity; (d) and (e) grade 3, spinal cord deformity.



Case Series by Ferreira et al. Sagittal T2-WI (A1-G1) and post-contrast T1-WI (A2-G2) show hyperintense "spindle shape" T2 signal in cases 1, 3, 4, 5 and 6 (A1, C1-F1) and focal-shaped in cases 2 and 7 (B1, G1).

Gadolinium contrast staining showed a "transverse band" in cases 1, 3, 4 and 7 (A2, C2, D2, G2); and "focal enhancement" in cases 2, 5 and 6 (B2, E2, F2).



- Ferreira et al, in the axial plane.
- Gadolinium contrast staining T1-WI showing "circumferential enhancement" in cases 1 and 4-7 (a, d, e, f, g), "focal shape" in case 2 (b), and diffuse in case 3 (c).

Management



- Treatment of CSM is primarily based on the severity and dynamics of the disease. Surgery is beneficial for patients with rapidly progressive or functionally severe disease (mJOA < 12).
- However, in practice, there is no consensus on how to manage patients with mild to moderate CSM (mJOA ≥ 12).

Non-operative



- Conservative treatment of CSM often includes neck immobilization, pharmacological treatment, lifestyle changes and physical modalities
- Kadanka et al compared conservative and surgical treatment of mild and moderate, non-progressive, and slowly progressive CSM (mJOA score >12) in 68 patients over 3 tahun → 3-year follow-up showed no superiority of surgery over conservative treatment

Operative



- Law et al. found that the progression of symptoms, the presence of myelopathy over 6 months, a compression ratio of about 0.4
- The decision to undergo surgery depends on many factors, including the duration of symptoms, the degree of spinal cord dysfunction, the patient's general condition, the degree of worsening, and radiographic findings.
- The main objectives of operative measures are:
 - to relieve spinal cord compression related microcirculation
 - restore cervical alignment
 - ensure long-term mechanical stability

Schematic Flowchart of Decision Making for CSM Surgery by Emery et al (2015)

CSM – Decision Making Factors						Choice of Approach
Sagittal Alignment		Kyphotic		Fixed		Anterior
				Flexible		Anterior or Posterior with Arthrodesis
		Neutral/Lordotic				Posterior (Laminoplasty) > Anterior
Number of Levels		3 or More				Posterior (Laminoplasty) > Anterior
		Less Than 3				Anterior > Posterior
Shape of Pathoanatomy		Hill shaped				Anterior > Posterior
		Flat				Posterior > Anterior
Age and Comorbidities		Elderly, Substantial				Posterior > Anterior
		Healthier				Anterior or Posterior
Pre-Operative Pain Levels		Moderate to High				Anterior or Posterior with Arthrodesis
		None to Low				Posterior (Laminoplasty) or Anterior
Instability/Subluxation		Yes				Anterior or Posterior with Arthrodesis
		No				Laminoplasty or Anterior

Choices of Operation Technique on CSM

Approach	Indications	Contraindications	Advantages	Disadvantages
ACDF	• Anterior pathology • Kyphosis • < 3 Levels	• Chin on chest deformity • Aberrant vertebral artery • Previous iatrogenic laryngeal nerve injury on contralateral side	• Less postoperative pain • Lower infection rate • Ability to correct kyphosis	• Bone graft complication • Swallowing difficulty or hoarseness • Dysphagia • Esophagus perforation • Pseudarthrosis
Corpectomy	• Circumferential decompression of the ventral cervical spinal cord	• Severe osteoporosis • Multilevel (> 3) • Chin on chest deformity • Aberrant vertebral artery • Previous iatrogenic laryngeal nerve injury on contralateral side	• More extensive decompression • Fewer graft surfaces to fuse • Can be combined with ACDF	In addition to complications of ACDF: • Vertebral artery injury • Dural tear • Adjacent segment degeneration • Greater blood loss • Longer operative time
Laminectomy alone	• Posterior pathology • Neutral to lordosis	• Inability to tolerate prone position • Active posterior infection • Chin on chest deformity • Cervical kyphosis	• Direct approach	• Delayed postoperative kyphosis • C5 radiculopathy • Dural tear
Laminectomy and fusion	• Posterior pathology • Multilevel CSM	• Inability to tolerate prone position • Active posterior infection • Shin on chest deformity • Cervical kyphosis	• Multilevel stabilization • More expansive decompression while providing stabilization via fusion	• Potential misplaced screws • C5 palsy • Vertebral artery injury • Dural tear • Infection
Laminoplasty	• “Tissue-sparing” alternative for spinal cord compression	• Inability to tolerate prone position • Active posterior infection • Chin on chest deformity • Kyphosis • Cervical instability	• Posterior elements preserved	• Limited posterior decompression • Late instability • Inconsistent relief of neck pain • C5 injury • Neck pain • Reduced range of motion • New-onset kyphosis
Combined ACDF and laminectomy with fusion	• Significant focal kyphosis • Posterior pathology • Multilevel decompression • Instability	• Inability to tolerate prone position • Active posterior infection • Previous irradiation to posterior neck	• Increased stabilization • Increased decompression	As above, and in addition: • More difficult technique • Longer operative time • Often requires staging

ACDF, anterior cervical discectomy and fusion.



OUTCOME

- Kadaňka et al assessed the 10-year outcomes of conservative vs surgical treatment of CSM patients. The study reported that on average, no significant differences in outcomes were found
- Sherief et al found that the overall complication rate of surgically treated CSM patients was 18.2%. This study found that the greater the preoperative disability, the greater the postoperative disability can be.
- Zika et al studied 36 surgically treated CSM patients at 1, 3, 12 months and 5 years postoperatively. → There was no correlation between gender, BMI, symptom duration, baseline mJOA, or degree of myelopathy at 12 months or 5 years after surgery.³⁴



RESEARCH METHODS

Methods



Study Type

The research method used in this research is descriptive research.
retrospective

Place and Time

The study used a sample of CSM patients at Prof. Soeharso Orthopedic Hospital during January 1, 2021 to December 2021, which was conducted for 3 months, namely June 2, 2022 to September 2022, through the hospital's medical records.

Sample and Population

All CSM patients of Prof. Soeharso Orthopedic Hospital from January 2021 to December 2021 were included in the population and sample of this study.



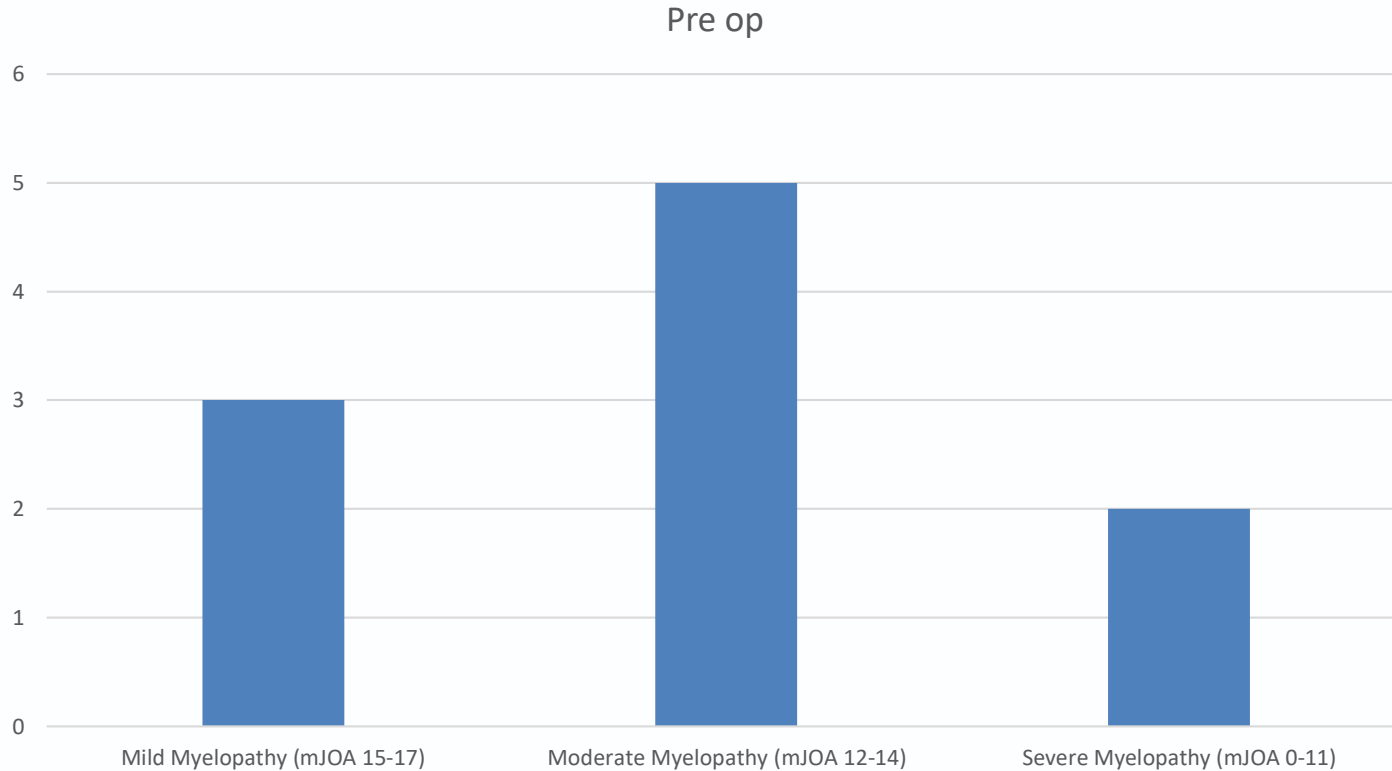
RESULTS

Demographic Data



Profile	Number of Patients	Percentage
Gender		
Men	4	40%
Female	6	60%
Age		
41-50	3	30%
51-60	5	50%
61-70	2	20%

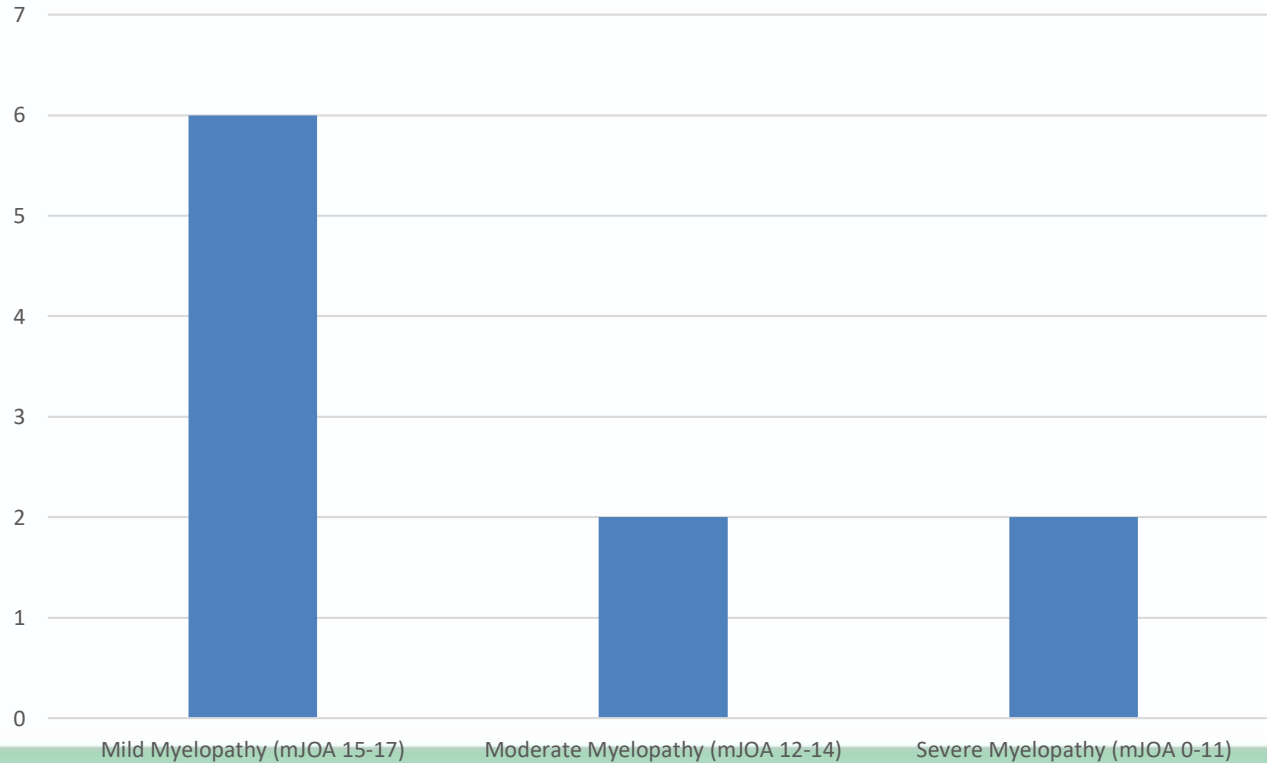
Pre op mJOA



Post op mJOA



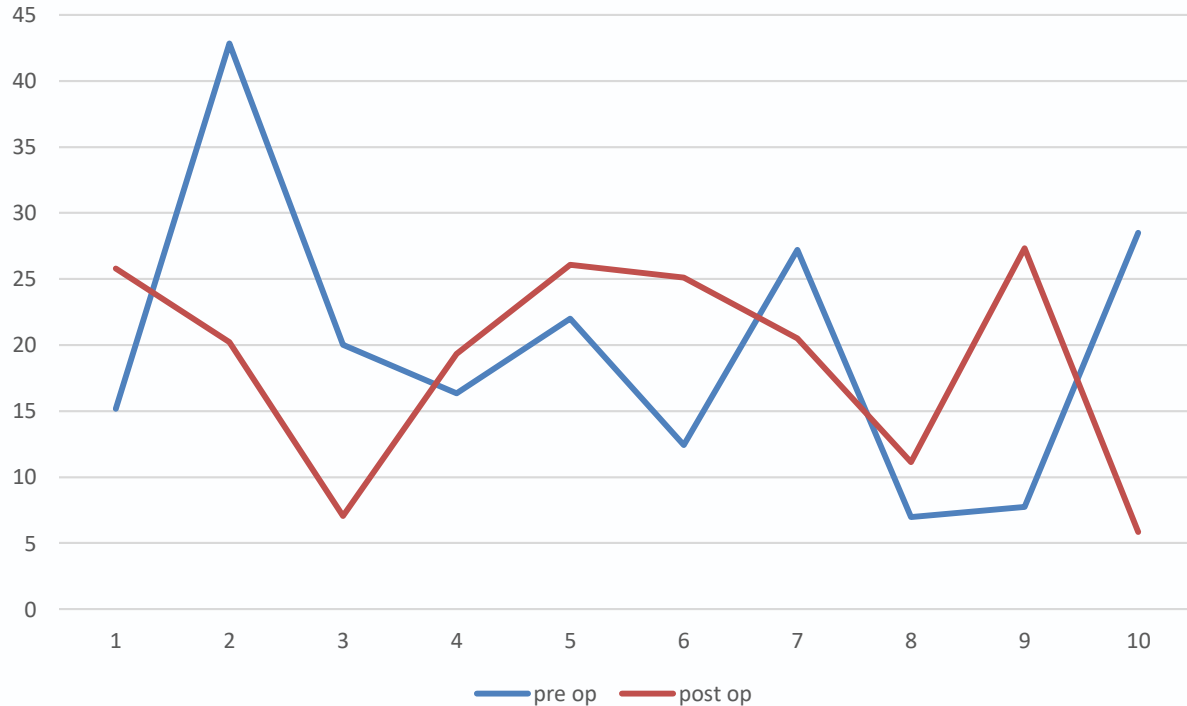
Post op

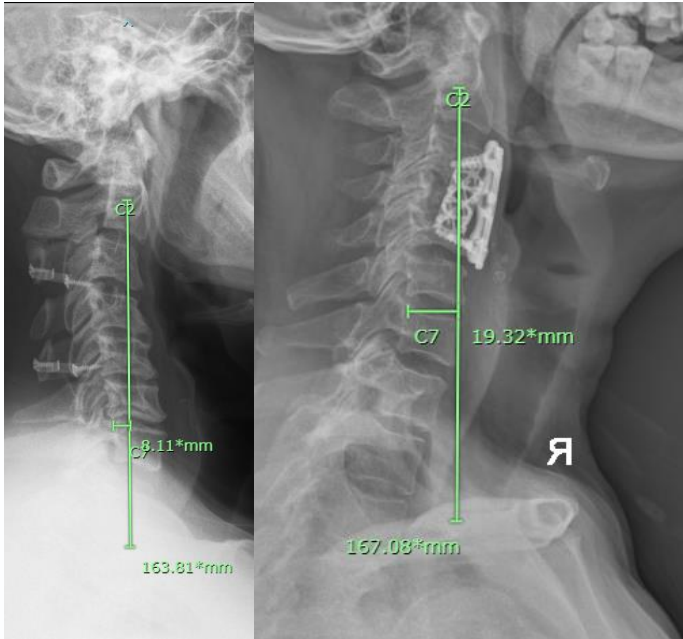


cSVA pre op and post op

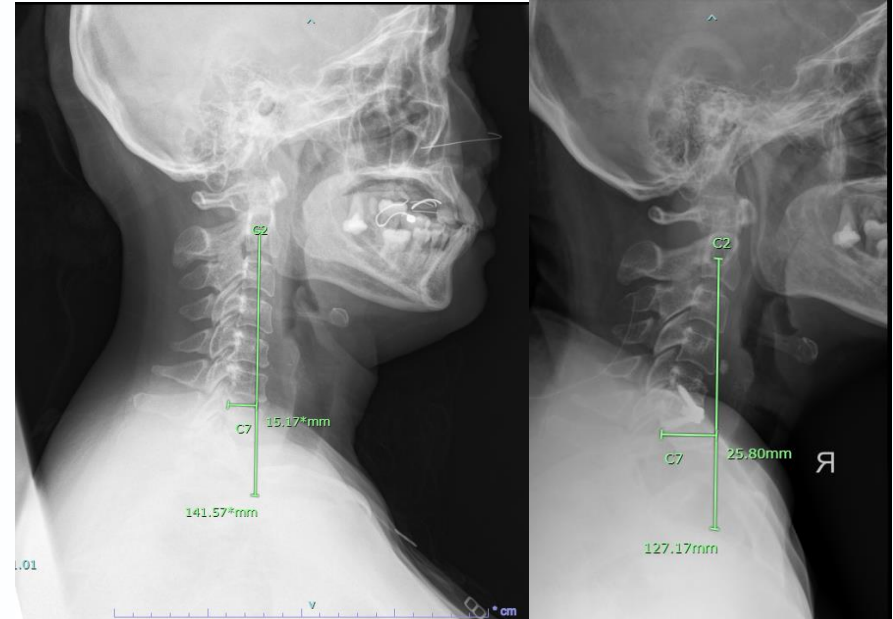


cSVA

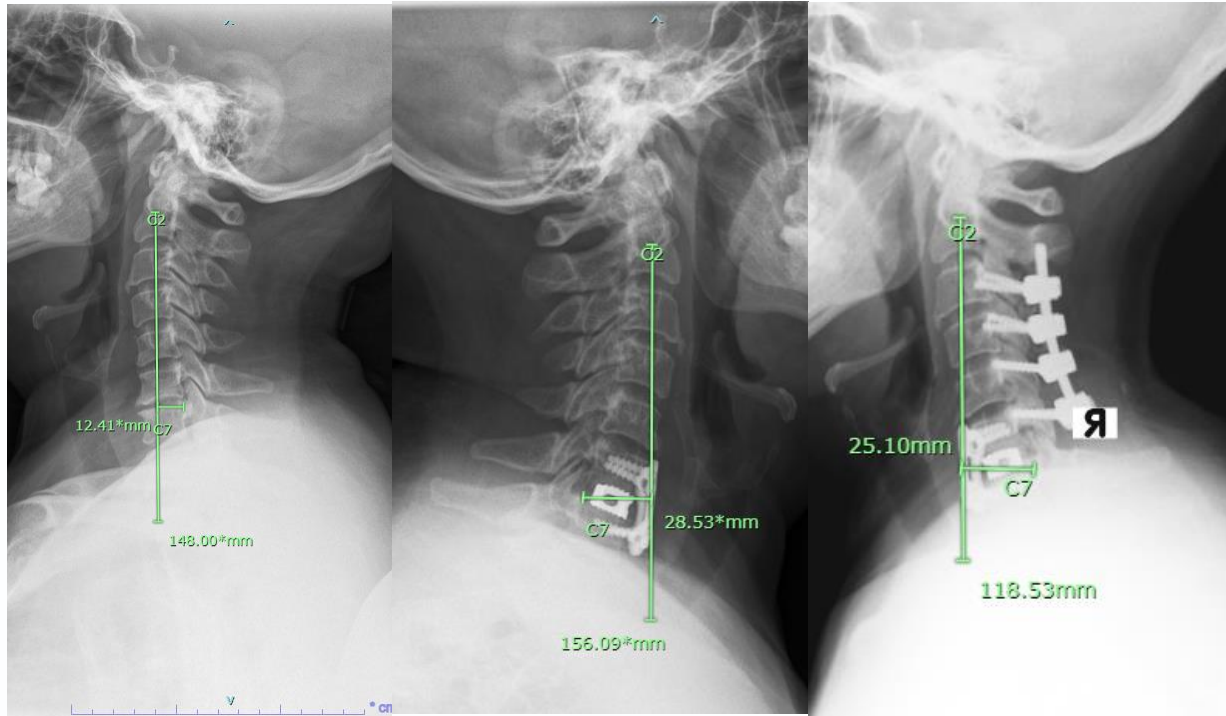




Patient Mr.M. cSVA pre op (left), post op (right)



Patient Mrs.H. cSVA pre op (left), post op (right)



Patient Mrs. N. cSVA pre op (left), post op (right)

Analytical Test



Relationship	r	p	Ket.
mJOA Pre-Op and cSVA Pre-Op	0.324	0.361	No
mJOA Post-Op and cSVA Post-Op	-0.100	0.783	No

The results of the Spearman correlation test between mJOA and cSVA before surgery obtained a correlation coefficient (r) of 0.324 and a significance value (p) of 0.361 ($p > 0.05$) so there was no significant correlation between mJOA and cSVA before surgery.

The results of the Spearman correlation test between mJOA and cSVA after surgery obtained a correlation coefficient (r) of -0.100 and a significance value (p) of 0.783 ($p > 0.05$) so there was no significant correlation between mJOA and cSVA after surgery.



mJOA	Mean	SD	p	Ket.
Pre-Op	12.20	3.01	0.004	Significant
Post-Op	14.50	3.17		

The results of the paired comparison test on mJOA conducted using the Wilcoxon test obtained a significance value (p) of 0.004 ($p < 0.05$) so that it was stated that there was a significant increase in mJOA after surgery.



DISCUSSION

Discussion



- 40% (n:4) male and 6 (60%) female patients

Wu et al	men have a higher incidence of CSM at all ages
Wang et al	higher males in CSM patients in China with a ratio of 3:1
Nagashima et al	male gender predominance in CSM patients below 80 years of age, while the percentage of men and women above 80 years of age was reported to be almost equal



- The youngest age range is 41-50 years while the oldest is 61-70 years.

Wang et al	highest in the age group 41-50, 51-60 years old
Sampath et al	mean age 48.7 +/- 12.03 years
Northover et al	The mean age of CSM diagnosis was 63.8 years.

Discussion



- Preoperative mJOA found 3 (30%) patients had mild myelopathy (mJOA 15-16), 5 (50%) patients had moderate myelopathy, and 2 (20%) patients had severe myelopathy.
- Postoperative mJOA with the results of 6 patients (60%) are mild, 2 patients (20%) moderate, and 2 patients (20%) severe.
- Wilcoxon test obtained a significance value (p) of 0.004 ($p < 0.05$) → significantly increased mJOA score after surgery.

Zika et al	mJOA increased from 10.8 ± 1.9 points preoperatively to 16.6 ± 2.2 points at 12 months
Pilato et al	An increase of 6 points or more in mJOA in 30 patients after surgery with a median of 16.43
Zaveri et al	Average increase in mJOA score from 11.01 to 16.4



Spearman correlation test results:

- mJOA and cSVA before surgery: correlation (r) of 0.324 and significance value (p) of 0.361 ($p > 0.05$) → there is no significant correlation between mJOA and cSVA before surgery.
- mJOA and cSVA after surgery obtained a correlation coefficient (r) of -0.100 and a significance value (p) of 0.783 ($p > 0.05$). → there is no significant correlation between mJOA and cSVA after surgery.

Lee et al	No clear correlation of cervical parameters to <i>clinical outcome</i> of patients
Evaniew et al	there is no significant association between cervical parameters and changes in quality of life, function, or health-related symptoms in their patients.

Conclusion



- CSM caused by degenerative changes in the spinal cord
- CSM can present with a variety of upper extremity-related symptoms such as gait disturbance, loss of manual function, muscle weakness, stiffness, or spastic gait.
- Operative management of CSM can improve the patient's mJOA score as a representative of *Functional Outcome* in patients

Thank You

